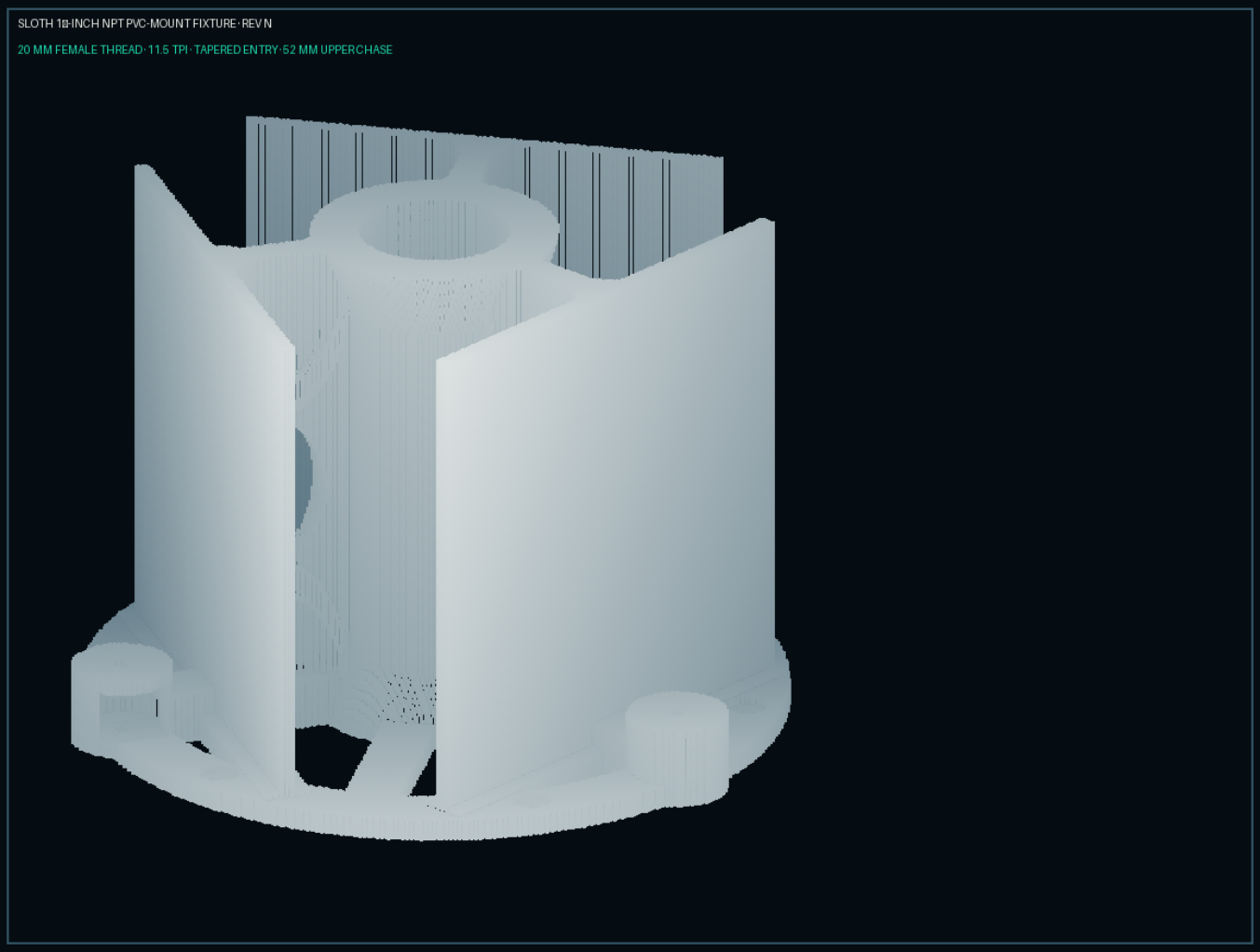
SLOTH HARDWARE / BUILD DOCUMENT

Three-Sector Passive Wi-Fi Antenna Fixture

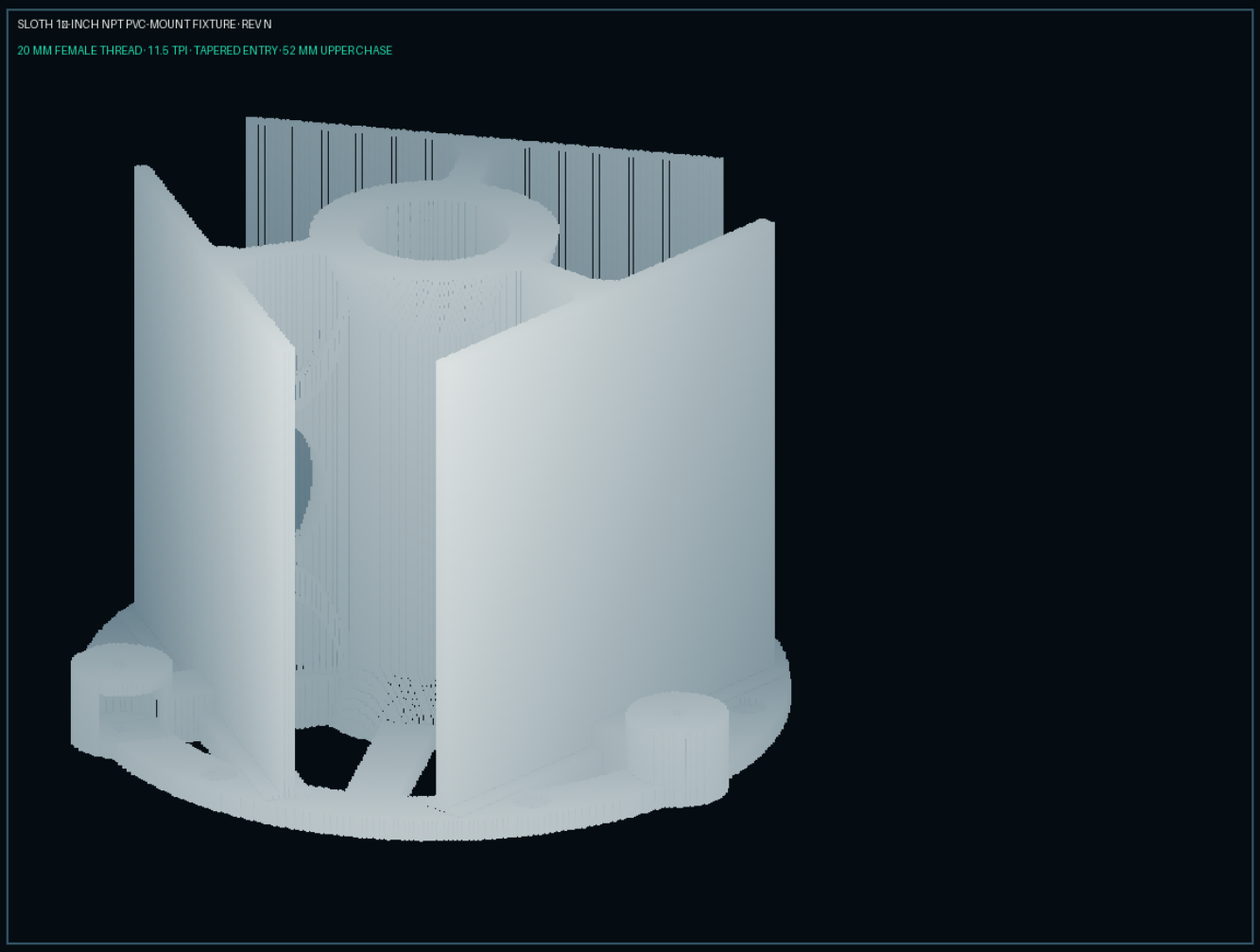
REV N - ABS / FDM / 1-1/2 INCH NPT PVC MOUNT



Complete printing, assembly, antenna installation, internal cable routing, reflector finishing, PVC mast mounting, and inspection guide.

Defensive passive receiving fixture. Not pressure-rated. Not intended for transmission or interference.

Finished result and major dimensions



FINISHED GEOMETRY

- Overall envelope: 248 x 240 x 173 mm. Compact 240 mm base ring with six recessed mounting locations.
- Three 5 mm planar reflector plates at 120-degree spacing; conductive foil is applied after printing.
- Hollow 52 mm center chase, three rounded structural cutaways, internal cable trunks, and serviceable antenna bosses.
- Bottom center: 20 mm-long female 1-1/2-11.5 NPT thread for a common PVC male adapter.

Print preparation - ABS FDM

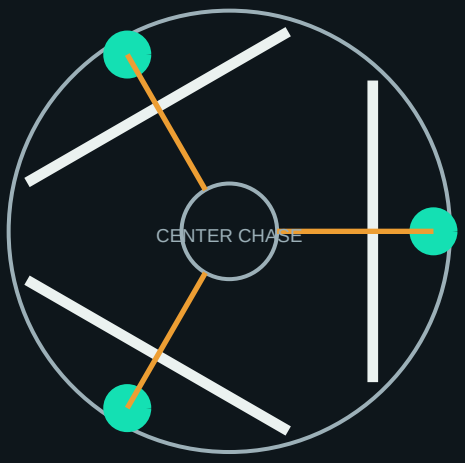
RECOMMENDED SETUP

- Fully enclosed printer; stable warm chamber.
- Minimum build area: 260 x 250 x 180 mm.
- 0.6 mm nozzle; 0.28-0.32 mm layers.
- 6-8 perimeter walls; 35-50% gyroid infill.
- Base-down orientation with 10-15 mm brim.
- Removable support beneath antenna decks if bridging is poor.
- Slow, enclosed cooldown before removal.

CRITICAL SLICER CHECKS

- Confirm all six mounting bores and top recesses are open.
- Inspect the 24 mm wrench bays and side tool ports.
- Verify every 10 x 10 mm cable trunk is continuous.
- Confirm the 52 mm center chase is open above the NPT collar.
- Use XY compensation based on an ABS calibration print.
- Print a lower-thread coupon before committing to the full part.

PRINT ORIENTATION



BASE DOWN

The wide, flat ring and spokes sit directly on the bed. Keep the reflector plates vertical. Do not rotate the model onto a reflector face; doing so weakens the plate-to-base joints and makes the internal trunks difficult to print.

ABS warning: the 240 mm ring can warp. Use a clean build surface, controlled chamber temperature, and gradual cooldown.

Parts, hardware, and tools

PARTS

- 1 x printed Rev N fixture
- 3 x Wi-Fi antenna elements
- 3 x SMA or RP-SMA bulkhead pigtails
- Conductive copper or aluminum foil
- 1 x 1-1/2-inch male NPT PVC adapter
- PVC mast or pipe compatible with the adapter
- 6 x M6 or suitable 1/4-inch base fasteners
- Washers and locknuts for the mounting surface

TOOLS

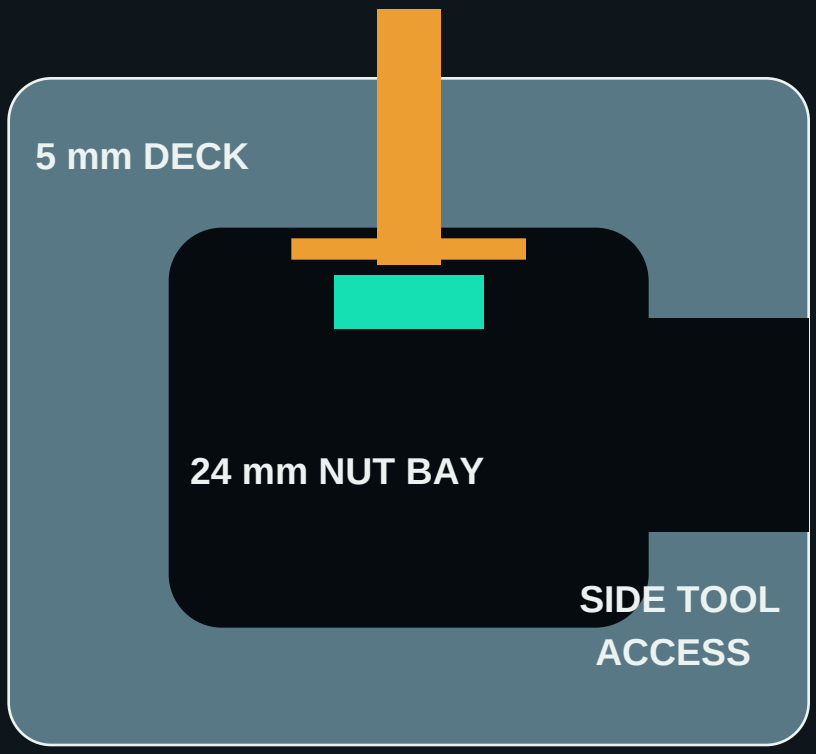
- Socket or nut driver under 22 mm OD
- Small open-end wrench or crowfoot
- 1-1/2-11.5 NPT tap for final fitting
- Deburring tool and fine file
- Fish tape or pull cord
- Multimeter for foil continuity
- Calipers
- ABS-compatible adhesive if secondary retention is required

BEFORE ASSEMBLY

- Deburr every opening without thinning the structural walls.
- Test each cable trunk with a pull cord before installing connectors.
- Dry-fit the exact PVC adapter. Chase the thread if it binds after one or two turns.
- Never use the printed connection as a pressure fitting or sole overhead restraint.

Install each antenna bulkhead

MOUNT CROSS-SECTION

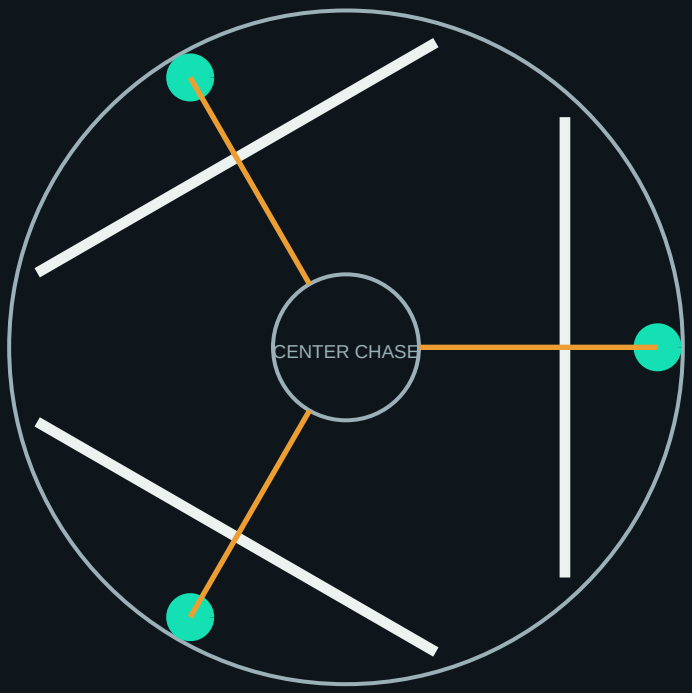


ASSEMBLY SEQUENCE

- Feed a pull cord from the center chase through the selected radial trunk.
- Insert the bulkhead connector downward through the 6.8 mm bore in the 5 mm mounting deck.
- Place the lower washer and nut inside the 24 mm wrench bay.
- Hold the upper connector stationary. Insert a socket, crowfoot, or small wrench through the side port and tighten the lower nut.
  Do not crush the ABS deck. The nut must be secure, not heavily torqued.
- Attach the coax to the pull cord and draw it inward through the trunk.

Route the three coax leads internally

TOP-VIEW ROUTING MAP

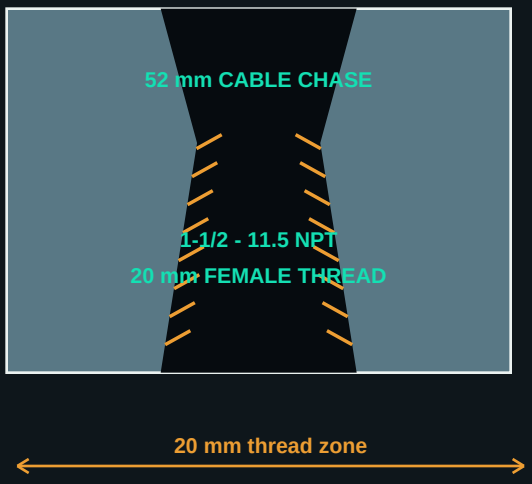


ROUTING RULES

- Keep bend radius within the coax manufacturer's recommendation.
- Pull one cable at a time and label it by sector before it reaches the center.
- The final short riser lifts each cable above the NPT thread before it enters the center chase.
- Add a soft grommet or spiral wrap where cables exit the center. Do not allow coax to rub against sharp printed edges.
- Leave a service loop, but keep cable and connectors clear of the foil reflector faces.

Mount the fixture to a PVC mast

THREAD SECTION



20 mm thread zone

PVC ATTACHMENT

- Use a 1-1/2-inch male NPT PVC adapter; NPT size is nominal, not the measured thread diameter.
- Deburr the entry and test by hand. The adapter should start squarely without cross-threading.
- Chase the printed thread with a 1-1/2-11.5 NPT tap for repeatable fit.
- Thread the adapter in by hand, then apply only modest tool torque.
- Use a secondary restraint for outdoor, moving, overhead, or safety-critical mounting.

IMPORTANT LIMITATIONS

- This printed NPT connection is a mechanical interface, not a pressure-rated plumbing joint.
- ABS and PVC can split under excessive tapered-thread engagement. Do not force the adapter to bottom out.
- Thermal cycling and UV exposure can loosen or weaken printed parts. Inspect regularly and use UV protection outdoors.

Apply reflector foil

REFLECTOR FINISHING

- Use the outward planar face of each 5 mm reflector plate.
- Clean the ABS with a compatible residue-free cleaner and allow it to dry.
- Cut foil slightly undersize so it cannot wrap into cable openings or contact connector hardware.
- Apply from the center outward with a soft squeegee to avoid wrinkles and air pockets.
- Use conductive tape at seams when multiple pieces are required.
- Verify electrical continuity across each reflector with a multimeter.
- Keep all three reflector treatments consistent so sector response remains comparable.

RF AND SAFETY NOTES

- The fixture is designed as a passive receiving structure for defensive Wi-Fi observation.
- Do not use it to transmit, jam, deauthenticate, or interfere with networks.
- The reflector geometry has not been calibrated as a certified antenna system. Validate sector response using legal, controlled test signals.

Final inspection and commissioning

MECHANICAL CHECKLIST

- No layer separation, warping, or cracks.
- All reflector plates are vertical and rigid.
- All base fasteners use washers and are snug.
- PVC adapter is square and not over-tightened.
- Antenna nuts are secure and still serviceable.
- Wrench bays and drain paths are clear.
- A secondary restraint is installed where required.

CABLE AND RF CHECKLIST

- All three coax leads are labeled.
- No cable is pinched or sharply bent.
- Center exit has edge protection and strain relief.
- Bulkhead connector center pins are undamaged.
- Reflector foil is smooth and continuous.
- Foil does not contact connector hardware.
- Passive scan confirms all three sectors are operational.

DOCUMENTATION AND MAINTENANCE

Record the printer, ABS material, slicer profile, thread compensation, antenna model, coax type, mounting hardware, and installation date. Reinspect after the first thermal cycle, after transport, and periodically during outdoor service.

Reference dimensions and source files

KEY DIMENSIONS

- Part envelope: 248 x 240 x 173 mm
- Base ring: 240 mm OD, 200 mm ID, 8 mm thick
- Reflector plates: 5 mm thick
- Antenna boss: 34 mm OD x 28 mm high
- Bulkhead bore: 6.8 mm
- Wrench bay: 24 mm diameter x 18 mm high
- Center chase: 52 mm diameter
- Female thread: 1-1/2-11.5 NPT, 20 mm long
- Mounting pattern: six 7.2 mm holes on a 216 mm bolt circle

STANDARDS AND FITTING REFERENCES

- ASME B1.20.1 - Pipe Threads, General Purpose, Inch
- Charlotte Pipe Schedule 80 PVC fitting catalog - 1-1/2-inch FPT/MPT adapters
- Parker NPT reference - 60-degree flank and 1-1/2-11.5 designation

Final fit is controlled by the exact printer and fitting. Use the supplied OpenSCAD source for dimensional adjustment and verify with the physical adapter before deployment.